

1. What is a Robot?

Core Definition: An electromechanical system characterized by three essential traits:

- **Reprogrammable** Flexible software control
- **Multifunctional** Versatile multi-task capability
- **Sensible** Real-time environment awareness

Key Terminology:

- **Robot:** Physical mechanism performing human tasks autonomously or via teleoperation.
- **Robotics:** Science of mechanical design, kinematics, control & software.
- **Telerobotics:** Remote-controlled operation in extreme/hazardous environments.

2. Automation vs. Robots

Fixed Automation

- **Single dedicated task**
- Maximum speed & efficiency
- Rigid; high retooling cost
- Ex: Bottling line, dishwasher

Flexible Robots

- **Diverse multiple tasks**
- Reprogrammable motions
- High mechanical flexibility
- Ex: 6-DOF arms, AGVs, CNCs

3. Asimov's 3 Laws of Robotics

- 1 **Safety First:** May not injure a human or, through inaction, allow harm.
- 2 **Obedience:** Must obey human orders, **except** when conflicting with Law 1.
- 3 **Self-Defense:** Must protect own existence, unless conflicting with Laws 1 or 2.

4. Core Hardware Architecture

- Sensors** **Data Input:** Internal state (encoders, resolvers) & external world (cameras, LIDAR, tactile/touch).
- Controller** **The 'Brain':** Executes trajectories, computes kinematics & closes real-time feedback loops.
- Actuators** **The 'Muscles':** Convert electrical/fluid energy into physical movement (servos, steppers, pneumatics).

5. Robot Types & Applications

Mechanical Architectures:

- **Manipulators:** Serial arms (Articulated, SCARA, Cartesian).
- **Mobile & Wheeled:** Differential drive, Omni, AGVs.
- **Legged:** Bipedal, quadrupeds, hexapods.
- **Autonomous Vehicles:** AUVs (Underwater), UAVs (Aerial drones).

Primary Drivers (3 D's / U's):

- Dangerous** Nuclear cleanup, bomb disposal.
- Dull / Repetitive** High-speed assembly, spot welding.
- Dirty / Unwanted** Sewer inspection, toilet cleaning.

6. Robot Programming Methods

ON-LINE METHODS (At the Robot)

- **Teach Pendant:** Handheld button box. Jog arm to waypoints & save; controller calculates PTP paths.
- **Lead-Through:** Manually guide arm by hand. Records continuous stream (60–80 pts/sec; high memory).

OFF-LINE METHODS (Remote / Code)

- **Programming Languages:** Text code (AML, VAL, RobotStudio) without halting factory line.
- **Task-Level:** High-level goals ("pick item A") auto-compiled into trajectory commands.

7. Performance Metrics

- **Working Volume:** 3D envelope reachable by end-effector.
- **Speed & Accel:** Motion rate; trade-off with precision & payload.
- **Resolution:** Smallest commandable incremental step.
- **Accuracy:** Closeness between **commanded target** and **actual reached position** ($|x_{\text{target}} - x_{\text{actual}}|$).
- **Repeatability:** Ability to return to the **exact same point** over repeated cycles (σ).

HIGH-YIELD EXAM DISTINCTION

Accuracy vs. Repeatability: A robot can have **High Repeatability** (hits the identical spot 100 times in a row) but **Low Accuracy** (that spot is systematically 2 cm off target)!

1. Sensing & Transduction

- **Sensing:** Collecting information about the world.
- **Sensor:** Device mapping physical/chemical phenomena to quantitative signals.
- **Transduction Principle:** Converting one energy form into another.

Sensor Type	Transduction Mechanism
Thermistor	Temp → Resistance (R)
Photodiode	Light → Current (I)
Pyroelectric	Thermal rad. → Voltage (V)
Humidity	Moisture → Capacitance (C)
LVDT	Position → Inductance (L)
Microphone	Sound press. → Elect. signal

2. Sensor Classifications

State Focus:

- **Proprioceptive** (Internal): Measures internal parameters (joint angle, wheel position, battery, tachometers/encoders/accelerometers).
- **Exteroceptive** (External): Environment & objects (proximity, vision, range).

Energy Interaction:

- **Active:** Emits energy & reads reflection (Sonar, Radar, Modulated IR, LiDAR).
- **Passive:** Receives ambient energy only (Cameras, Pyroelectric, LDR).

Contact & Modality:

- **Contact / Non-contact:** Touch (bumpers) vs. remote (optical, ultrasonic).
- **Visual / Non-visual:** Vision cameras vs. non-optical sensors.

3. Contact & Resistive Sensors

A. FEELERS (TACTILE/CONTACT)

- **Whiskers:** Piano wire in metal hoop; deflection closes circuit; binary (0/1) output.
- **Bumpers & Guards:** Frame on microswitches; collision triggers switch; binary (0/1) output.

B. RESISTIVE (VARIABLE R)

- **Bend Sensors:** Flexible strips where R increases as bent ($10\text{ k}\Omega - 35\text{ k}\Omega$). *Uses:* Joint angles, wall-following, loads.
- **Potentiometers:** Linear/rotational variable resistors tracking sliding parts or rotating joint shafts.
- **Photocells (CdS / LDR):** Resistance shifts with light; highly non-linear, ideal for light tracking.

4. Capacitive vs. Inductive Sensors

Capacitive (All Materials)

- Measures change in **dielectric**
- Detects **metals & non-metals** (liquids, plastics, powders)
- Senses through glass/plastic
- Sensitive to dust & humidity

APPLICATIONS

- **Level Detection:** Liquid/grains
- **Proximity:** Assembly lines
- **Touch Sensing:** Touchscreens

Inductive (Metals Only)

- Emits AC magnetic field; senses **eddy currents**
- Detects **metals ONLY**
- Max on **ferrous** (iron/steel)
- Immune to oil, dust & water

APPLICATIONS

- **Position:** Metal part automation
- **Speed:** Rotating gears/shafts
- **Metal Detection:** Safety

5. Infrared (IR) Sensor Types

Type	Principle & Trade-offs	Applications
Intensity-Based Reflective	• Measures reflected light • Pro Cheap, simple • Con Ambient & color sensitive	• Line-following • Wall-tracking • Break-beams
Modulated IR Proximity	• Flashes at $32 - 45\text{ kHz}$ • Receiver demodulates signal • Pro Immune to ambient light	• Proximity sensing • Remote controls • Obstacle avoidance
IR Triangulation PSD Ranger	• Emitter + lens + PSD • Angle of return gives range • Pro Color & light immune	• Sharp GP2D02 ($10 - 80\text{ cm}$) • Distance map

6. Ultrasonic Sensors & Noise

- **Principle:** Ultrasound burst ($\approx 50\text{ kHz}$). Time-of-Flight (t):

$$D = v \cdot t \implies d = \frac{v \cdot t}{2} \quad (v \approx 340\text{ m/s in air})$$

- **Limitations:**
 - **Bearing Uncertainty:** Wide beam spread (opening arc $\approx 30^\circ$).
 - **Speed Latency:** Sound is slow (e.g. 30 m round-trip takes $\approx 200\text{ ms}$).
- **Noise Issues:** Acoustic noise & sensor crosstalk.

7. Laser Range Finders (LiDAR)

- **Performance:** Long range ($2\text{ m} - 500\text{ m}$), high resolution (10 mm), fast scans ($13 - 40\text{ ms}$).
- **Robustness:** Narrow beam; immune to ambient light & specular reflections.

HIGH-YIELD EXAM COMPARISON

Inductive vs. Capacitive: Inductive senses **metals only**; Capacitive senses **all materials** (dielectric shift).
Ultrasonic vs. LiDAR: Sonar has wide **bearing uncertainty** ($\approx 30^\circ$) & latency; LiDAR provides millimeter angular precision.

1. Agency & Marr's 3-Level Framework

- **Agency:** Capacity of an entity to act intentionally, make choices, and exert control over actions and outcomes.
- **David Marr's Framework:** Decomposes information-processing systems into 3 empirical layers:
 1. **Level 1: Computational (Theory):** What is the agent doing & Why? Ultimate goal, mathematical theory & constraints.
 2. **Level 2: Algorithmic (Representation):** How is the task processed? Input/output representations, transformations & planning.
 3. **Level 3: Implementational (Physical):** How is it physically realized? Hardware/substrate (synapses/muscles vs. MCUs/actuators).

CRUCIAL EXAM FACT L1 & L2 are universal to all agents; differences between biology & robots exist **only at Level 3**.

2. Case Study: Mosquito vs. Robot

Marr Level	Mosquito (Bio)	Search & Rescue (Robot)
L1: Comp.	Locate warm host for blood meal	Locate human survivors in disaster
L2: Algo.	Gradient ascent on heat & CO ₂	Steer based on thermal & CO ₂ inputs
L3: Impl.	Olfactory pits, flight muscles	Thermopile, CO ₂ sensor, MCUs, wheels

3. Biological Inspiration Mappings

Animal / Task	L2: Algorithm	L3: Implementation
Bat (Avoidance)	Echo delay ranging	Ultrasonic sensors & CPU
Ant (Pathfind)	Ant Colony Optim. (ACO)	Digital pheromones/ maps
Bird (Flocking)	Reynolds Boids (Cohesion, Alignment, Separation)	UAV drones + local wireless communication
Frog (Grasp)	Motion track + reflex	High-speed camera + arm
Cuttlefish	Spatial pattern mapping	E-Skin & vision cameras

4. Definition of Robotic Behavior

- **Behavior:** Mathematical mapping of sensory inputs to motor action patterns to achieve a task.
- **Action-Perception Cycle:** Continuous loop where perception drives action, action alters environment, and environment feeds perception.
- **Gibson's Ecological Approach:** Perception exists solely to support action. **"The world is its own best representation."**
- **Affordances:** Directly perceivable action possibilities (e.g. **climbability, sitability**) without complex mental reconstruction.

Mathematical Formulations:

- **Stimulus-Response (S-R):**
$$\{B : S \rightarrow R\}$$
- **Schema Notation:**

$$B[S] = R$$

- $S[\text{Sensor Data}] \rightarrow \text{Percept}$ (filters raw noise into percept).
- $R[\text{Percept}] \rightarrow \text{Action}$ (maps percept to motor command).

5. Taxonomies of Behaviors

1. REFLEXIVE (S-R)

- Hardwired, direct neural/electronic connection for fastest execution.
- **Fixed-Action Pattern (FAP):** Response continues for **longer duration** than the stimulus (e.g. web spinning, courtship dance, nut burying, duckling imprinting).
- **Taxis:** Orienting reflex relative to stimulus (e.g. **phototaxis** turning towards light). *Can co-occur with FAP.*

2. REACTIVE (LEARNED / MUSCLE MEMORY)

- Learned through repetition and consolidated into automatic execution without conscious thought (e.g. riding bike, skiing).
- *Conscious overthinking degrades reactive performance.*

3. CONSCIOUS (DELIBERATIVE)

- Goal-directed, high cognition, planning, reasoning, puzzle solving.

6. Schema Theory in Robotics (OOP)

- **Schema Theory:** Translates animal behavior into reusable **Object-Oriented Programming (OOP)** classes.
- Foundation of the **Reactive Layer** in hybrid architectures.
- **Anatomy of a Schema Class:**
 1. **Schema-Specific Data** → Member Variables (models, states).
 2. **Procedural Knowledge** → Methods (computational algorithms).
- **Schema Instantiation (SI):** Generic template class parameterized with runtime values (e.g. bike seat height → active SI object).
- Primitive behavior = **1 Perceptual Schema** + **1 Motor Schema**.

7. Vector Fields & Frog Snapping

- **Frog Algorithmic Schemas:**
 - **Feed Schema:** Small moving blob → Snap tongue.
 - **Flee Schema:** Large moving blob → Runaway jump.
- **Level 3 Vector Fields:**
 - Stimulus mapped as vector (m, θ) .
 - **Distance-Intensity Scaling:** Weaker stimulus (far away fly) generates a **harder snap vector**.
 - **Vector Summation (Σ):** Left eye vector (v_L) + Right eye vector (v_R)
 $\Rightarrow V_{\text{resultant}} = v_L + v_R$ for optimal tongue trajectory.

8. Innate Releasing Mechanisms (IRM)

- **Releasers as Boolean Latches:**
 - **PRESENT** (true) → Behavior executes.
 - **NOT_PRESENT** (false) → Gated entirely (no default output).
- **Compound Releasers:** Condition combines **External Stimuli** (food sensed) + **Internal State / Motivation** (hunger level / battery).
- **Implicit Chaining:** Linking schemas via releaser states allows complex sequences to emerge without a centralized planner.

EXAM HIGHLIGHTS: LECTURE 03

Marr's Level 3: Only hardware/biology separates humans & robots; L1/ L2 are universal.
FAP vs. Taxis: FAP action outlasts the stimulus duration; Taxis is directional steering.
IRM Gating: When releaser is false, behavior is shut off with **no default output**.

1. LEGO Mindstorms EV3 Architecture

- **Processing Unit:** ARM9 @ 300 MHz, 16 MB Flash, 64 MB RAM.
- **Operating System:** Embedded Linux-based OS.
- **Sensor Ports (1-4):** 4 Input ports supporting Analog & high-speed Digital UART up to 460.8 kbit/s.
- **Motor Ports (A-D):** 4 Output ports with built-in optical rotation encoders for position/speed feedback.
- **I/O & Expansion:**
 - High-speed USB (480 Mbit/s) + Host daisy-chaining (up to 3 levels), WiFi dongle, & USB storage.
 - Micro SD-card slot (supports up to 32 GB).
- **Display & UI:** 178 × 128 monochrome LCD matrix, 6 backlit navigation/debugging buttons.
- **Wireless Comms:** Bluetooth v2.1 + EDR, Apple/Android smart device protocol.

2. Platform Generations & Modules

Platform	CPU / OS	Key Ports & Features
RCX (1998)	Hitachi H8	3 In / 3 Out; Optical IR tower comms
NXT (2006)	ARM7 + AVR	4 In / 3 Out; Bluetooth, 100 × 64 LCD
EV3 (2013)	ARM9 Linux	4 In / 4 Out; Daisy-chain, USB Host, SD

Sensor & Actuator Modules:

- **Motors:** Large (high torque) & Medium (high RPM) with integrated tacho-encoders.
- **Sensors:** Touch (binary limit), Color (ambient/reflected/RGB), IR Seeker/Beacon, Ultrasonic.

3. Step 1: Avoid Collision (Reflex)

PURE REFLEXIVE BEHAVIOR

- **Objective:** Move continuously; instantly back up and pivot when a physical impact occurs.

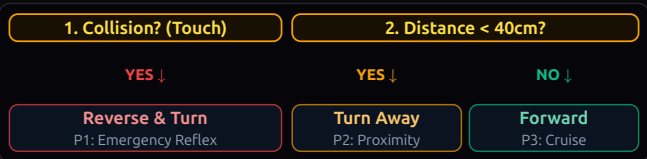


- **Collision == True:** Reverse (−30, 2 rot) → steer turn (−45/30, 2 rot).
- Direct hardware-to-motor S-R loop without state memory.

4. Step 2: Collision + Obstacle Avoidance

HIERARCHICAL PRIORITY ARBITRATION

- Combines contact (Touch) and non-contact (Proximity/Force) into a 2-tier reactive hierarchy:

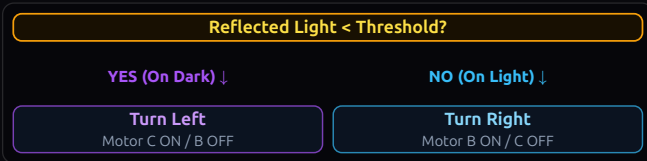


- **P1 (Touch):** Emergency physical recovery (overrides all).
- **P2 (Proximity):** Smooth steering turn away before contact.
- **P3 (Default):** Drive straight ahead.

5. Step 3a: 1-Sensor Line Following

BANG-BANG / EDGE TRACKING

- Tracks the **boundary edge** between light and dark surfaces using 1 photo/color sensor.



- $Ref < Th$ (On black) → Steer outside (Turn Left).
- $Ref \geq Th$ (On white) → Steer inside (Turn Right).
- **Control Characteristic:** 2-state discrete bang-bang controller; causes high-frequency zigzag oscillation along line edge.

6. Reactive Control Summary

Stage	Sensor Used	Behavior Pattern
Step 1	Touch Switch	Emergency reflex backoff
Step 2	Touch + IR/Dist	Layered priority avoidance
Step 3a	Color/Light	1-Sensor edge tracking

EXAM HIGHLIGHTS: LECTURE 04

- **EV3 vs NXT/RCX:** EV3 introduced Linux OS, ARM9 @ 300MHz, daisy-chaining, USB host, & 4th motor port.
- **1-Sensor Line Following:** Always tracks an **edge** (boundary transition), never the centerline itself.
- **Priority Arbitration:** Safety/impact reflexes always take precedence over non-contact navigation.